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Introduction to the formal semantics of ODRL policies

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Policy Evaluation and Enforcement on the Web with ODRL
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My Short Bio

- Since September 2017, I am **Lecturer and Researcher** at the Institute of Digital Technologies for Communication, Faculty of Communication, Culture and Society at Università della Svizzera italiana (USI), Lugano, Switzerland
- My research is focused on the field of **Agents and Multiagent Systems** and **Semantic Web**
- Since 2007, I am member of the Steering Committee of the **COINE** (Coordination, Organization, Institutions and Norms in agent systems) Workshop series
- Since 2021, I am Co-Chair of the W3C ODRL (Open Digital Rights Language) Community Group. I coordinate the activities of the group that is defining the **semantics of ODRL**
- In June 2025 I co-organized the **OPAL 2025** Workshop "ODRL and beyond: practical applications and challenges for policy-base access and usage control." at the **ESWC2025** (June 1 and 2, 2025 Portoroz, Slovenia)
- In July 2025 I organized with Marina De Vos, Munindar P. Singh, and Leon van der Torre, the Dagstuhl Seminar "Policy Modeling and Reasoning in Sociotechnical Systems" (Jun 29 – Jul 04, 2025)
- In May 2026 I co-organized the **OPAL 2026** Workshop and the ESWC2026 tutorial: "Policy Evaluation and Enforcement on the Web with ODRL"

The fundamental Requirement #1 of all policy languages

- Policies are written by some actors A, usually the assigner or creditor of the policy, and shall be **complied** with by other actors B, usually the assignee or debtor of the policy
- A and B **should understand the policy in the same way**, otherwise **compliance** is not possible even if B wants to comply
- This is a fundamental requirement also for policy negotiation
- Only **formal semantics** guarantees this, it must be unambiguous, a natural language formulation is not enough
- ODRL 2.2 Recommendation [2] does not include the specification of a formal semantics
- There is a sub-group within the ODRL Community Group that is working on a proposal starting from **concrete examples** of **policy types** [1]

Another crucial Requirement #2: Task Independence

- Policies are used for a variety of **tasks** (with a variety of names):
  - Access control / enforcement
  - Auditing / monitoring / usage control
  - Negotiation / agreement
  - Explanations (why / why-not / how-to / what-if queries)
- Semantics should be **task-independent** (what, not how) and define **one** correctness criterion for all tasks.
- E.g. if what-if says “ok the action is permitted” then access control permits, and monitoring does not notify violations

What I will and will not presenting

- Definition of the **semantics of ODRL** based on *ODRL Information Model 2.2* [2] as it is specified in [1] (a “Model theoretic” declarative semantics based on simplified temporal Kripke structures (traces) is specified in [3])
- Covering the semantics of the main **rule types** of the language: **permissions, prohibitions, and obligations**
- Not covering history-dependent properties (e.g. count)
- Identifying ambiguities, **gaps**, and contradiction in the informal semantics [3]
- Discussing proposals for fixing problems [3]
- Providing examples to validate the semantics

Outline of the presentation

Compliance with Permissions, Prohibitions, and Obligations

In order to compute compliance we need to specify how an **ODRL Evaluator** has to determine:

- which Permissions, Prohibitions, and/or Obligations (collectively named Rules) are **active** on some action a in some state of the world. A Rule is said to be active if it is in effect;
- If an action *a* complies (or **is permitted by**) with a **Permission** in a given state of the world;
- If an action *a* complies with a **Prohibition** in a given state of the world, otherwise *a* is **prohibited** by the Prohibition;
- If an action *a* complies with an Obligation in a given state of the world, *a* is called the **fulfilling** action of the Obligation.

ODRL Evaluator

The ODRL Evaluator uses as **input**:

- a **Policy**;
- a formal description of the **State of the World** that lists the **events/actions** that occur and the **time** when they occur
- a formal description of an **Evaluation** Request, it represents a formal description of an **action** to be evaluated
- an optional parameter specifying the **Behavior** of the system.

The ODRL Evaluator produces as **output**:

- The **Evaluation Report**.

Permissions and their intuitive semantics

- See in [1] 4 examples of different types of permissions
- A **Permission** inside a **Policy** is characterized by:
  - An action class **α**, e.g. distribute, use, transfer; the action can be refined by "refinement"s **leftOperand operator rightOperand**
  - A "target" asset **ob**;
  - An "assignee" party **p**;
  - Constraints **leftOperand operator (op) rightOperand**
  - Other optional properties, e.g. assigner, **Duty/Conditions** (with action class **α2**, target **ob2**, assignee **p2**, refined by refinements, and constraints)
- Intuitive meaning from [1]: "A Permission **allows** an action, with all refinements **satisfied**, to be **exercised** on an Asset if all constraints are **satisfied** and if all duties are **fulfilled"**.

Semantics of Permissions (1)

- The ODRL Evaluator computes that the action **a** represented in the Evaluation Request **is permitted by a Permission** in the given State of the World **S** **if and only if all** the following conditions holds:
  1. The Permission is **active** on action a in state S (see next slide);
  2. The action a belongs to the permitted class of actions;
  3. The action's target is equal to the target of the Permission or it is contained in the collection of objects specified as target of the Permission;
  4. When the assignee of the permission is specified, the action's party is equal to the assignee of the Permission or it is contained in the collection of parties specified as assignee of the Permission;
  5. All the refinements (when they are specified) related to the regulated action, to the regulated target, and to the regulated assignee are satisfied by the action a, its target and its party.

Semantics of Permissions (2)

- The ODRL Evaluator computes that a Permission is **active** on some evaluated action ***a*** represented in the Evaluation Request and in some State of the World ***S* if and only if all** the following conditions holds:
    1. All the existing **Constraints** of the Permission are **satisfied** by the evaluated action ***a***; (for every constraint the value of the action's (or asset's) property leftOperand is op of the value denoted by the the rightOperand, e.g. "2017-12-19" It "2018-01-01")
    2. All the existing **Conditions** (duties) of the Permission are **fulfilled** or **inactive** in state **S** with respect to the action **a**. Duties are prerequisites.
- If one of those conditions is not met, the Permission is **inactive**.
- Assumption: the ODRL Evaluator is able to compute the satisfaction of constraints, this requires a mapping between the properties of the policy, the evaluated action and the state of the world that are managed independently

Semantics of Conditions of Permission (1)

"All the existing **Conditions** (duties) of the Permission are **fulfilled** or **inactive** in state **S** with respect to the action **a**" (the action in the Evaluation Request).

- A Condition (duty) is **active if and only if one** of the following conditions holds:
  1. The Condition (duty) has no Constraints;
  2. Every **Constraint** of the Condition (duty) is **satisfied** by the action **a** (the evaluated action ***a*** represented in the Evaluation Request)

  Otherwise, the Condition is **inactive.**

Semantics of Conditions of Permission (2)

- A **Condition** (duty) is **fulfilled** **if and only if** **all** the following conditions holds:
  1. The **Condition** (duty) is **active**;
  2. There exist an action **a2** in the **state of the world** that belongs the class of actions regulated by the **Condition** **α2**;
  3. Such an action **a2** satisfies all the **refinements** of the action regulated by the **Condition** **α2**;
  4. If a target object of the Condition is specified, the assets involved in action **a2** are exactly those specified in the target of the Condition and its refinements;
  5. If an assignee party of the Condition is specified, the parties that exercise action **a2** are exactly those specified in the assignee of the Condition and its refinements.
- An active Condition that is not fulfulle is **not-set**

Remark 1

- Alternative semantics for fulfilment of Condition (Duty)
  - (**current**) the action should be exercised **jointly** by all the specified assignees: e.g. *the contract shall be approved by all assignees*
  - (**alternative 1**) the action should be exercised by **any** of the specified assignees, e.g. *the payment shall be made by any of the assignees*
  - (**alternative 2**) **parallel**, each assignee should independently fulfil the coniditon/duty: e.g. *all assignees shall register*

ODRL does not specify which of them is the correct one, *it should be possible to choose the alternative!*

Remark 2

- The semantics of Duty used as **Condition** of a Permission is different from the semantics of an **Obligation** to perform an action:
  1. The **constraints** of a **Condition** are evaluated with respect to the action to be evaluated regarding the Permission
  2. The **constraints** of an **Obligation** are evaluated with respect to the obliged action
  3. An agent is not obliged to fulfil the **Condition** for obtaining a permit, they are free to choose without incurring a violation. If, on the other hand, an agent has an obligation and fails to fulfil it within the specified time, they are in violation.
- *It should be better to have two different classes: Obligation/Duty and Condition*

Remark 3

- Duties are **prerequisites** for getting an **active Permission**; their fulfilment must occur **before** the permission is granted;
- This version cannot model duties that must be fulfilled **after** the permitted action is actually performed, e.g. personal data shall be deleted within one month from the access to them;
- *The second meaning could be represented with an obligation activated by the performance of an action, e.g. the access to the data*

Remark 4

- In the previous semantics a **single fulfillment** of the duty is enough to perform **any number** of actions, e.g. After registering to a web site a user may access all the content of the web site.
- In other use cases could be useful to say that **each** action execution requires a different fulfillment of the duty; e.g. Pay per view policies
- *It should be better to distinguish these two meanings*

Remark 5

- In the ODRL specification [1]:
  “A Permission **allows** an action... if all duties are **fulfilled**”,
  but from the discussions within the Community Group, it emerged that if the **duty/condition** is **inactive**, it is not relevant for computing if the action is permitted or not!

References

[1] Fornara, Nicoletta, Víctor Rodríguez-Doncel, Beatriz Esteves, Simon Steyskal, Benedict Whittam Smith, and Yassir Sellami, Andrea Cimmino Arriaga. **ODRL Formal Semantics** (Draft Community Group Report) https://w3c.github.io/odrl/formal-semantics/.

[2] Iannella, Renato, and Serena Villata. **ODRL Information Model 2.2** (W3C Recommendation 15 February 2018). https://www.w3.org/TR/odrl-model/.

[3] Piero A. Bonatti, Nicoletta Fornara, and Andreas Harth. **Towards a Formal Semantics of the Open Digital Rights Language (ODRL 2.2)**. ODRL and Beyond: Practical Applications and Challenges for Policy-Based Access and Usage Control (OPAL 2025), Co-Located with the Extended Semantic Web Conference 2025 (ESWC 2025), 2025. https://ceur-ws.org/Vol-3977/OPAL2025-4.pdf .

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